XRPL Validator — Daily Engineering Report · 2026-07-19

Period: 2026-07-19 00:00 to 24:00 US Eastern (2026-07-19 04:00 to 2026-07-20 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-19): 1.00000 (0 missed of 21,662). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(xrpl-ledger): order-book quality keylet + OfferCreate placement — base camp** (`150c432`, xrpl-validator) The book quality directory key was the last missing keylet family: - book_base = SHA512Half(0x0042 'B' + pays_cur + gets_cur + pays_iss + gets_iss); quality dir = base[0..24] + quality_be64 - offer_quality mirrors rippled getRate under the STNumber switchover: normalize both mantissas to [1e15,1e16), truncating muldiv at 10^17, +5, then ONE round-half-even pass over the excess tail (Number canonicalize). Locked …
- **feat(xrpl-ledger): integer offer-crossing engine — the summit assault** (`7ee6dcd`, xrpl-validator) Replaces the f64 crossing stub with a real integer engine: - SHAMap::keys_with_prefix (nibble-radix descent) + Sandbox merged view — native book enumeration: walk a book's quality dirs ascending from best quality, stop past the taker's threshold (swapped-args offer_quality) - (mantissa u128, exponent i32) arithmetic with directed rounding at fill boundaries (taker pays UP, receives DOWN — Quality::ceil_in/out …
- **fix(xrpl-ledger): cancel-and-replace + funded-check starvation — OfferCreate 64%** (`f5b87b2`, xrpl-validator) Two data-diagnosed fixes to the crossing engine's first corpus run: - OfferSequence on OfferCreate cancels the named prior offer BEFORE crossing (rippled cancel-and-replace; the '126 crossing misses' were the taker's own old offers — replaced, not crossed) - differential read-set loads the taker's gets-side trust line: a pure placement never has it in mainnet meta, so available() starved and threw 220 false …
- **feat(xrpl-ledger): IOU + cross-currency payment engine — second summit** (`e6efe32`, xrpl-validator) Payment was XRP-only (preflight returned tecUNSUPPORTED for IOU; preclaim demanded an XRP balance even for cross-currency payments BUYING XRP). Segmentation showed the corpus is dominated by arb-probe

partial payments (XRP-deliver, IOU SendMax, tfPartialPayment) that mainnet fails tecPATH_DRY fee-only — natives just needed to attempt the path and agree.

- **feat(xrpl-ledger): CheckCash via shared transfer engine — IOU checks + tecPATH_PARTIAL semantics** (`a4dc8d7`, xrpl-validator) CheckCash now routes through the offer.rs transfer helpers (leg_of/available/move_leg) instead of hand-rolled XRP-drops balance edits: IOU checks cash over trust lines, issuer self-cover is unlimited, and writer shortfall fails the rippled way (tecPATH_PARTIAL, fee-only) instead of tecUNFUNDED. Check deletion now unlinks BOTH owner directories (OwnerNode + DestinationNode hints) and decrements both owner counts. …
- **feat(xrpl-ledger): CheckCash via transfer engine + tec classification — corpus 90%** (`02508d9`, xrpl-validator)
- CheckCash rides the shared engine: SendMax fixes the currency, writer shortfall (or over-asking) fails the rippled way (tecPATH_PARTIAL, fee-only), success moves the leg + deletes the check + unlinks BOTH owner directories (OwnerNode/DestinationNode hints) + OwnerCounts. CheckCash 0/30 -> 30/30. - probe read-set loads the CheckID object (a fee-only mainnet meta never contains it -> native said tecNO_ENTRY where …
- **fix(xrpl-ledger): path payments — quality gate only under tfLimitQuality, DeliverMin floor, book pre-state — corpus 90.3%** (`28f0cb6`, xrpl-validator) Two root causes behind the 62 tecPATH_DRY divergences: - Engine: apply_path_payment imposed the SendMax/Amount ratio as a per-offer quality bound unconditionally. rippled only does that under tfLimitQuality; arb-style conversions use a sentinel-max Amount whose implied ratio reads every book as dry. Threshold is now unlimited without the flag (aggregate SendMax/Amount bounds still hold), and DeliverMin is enforced …
- **feat(xrpl-ledger): AMM swap engine — corpus 94.1%, the third summit** (`18d4a00`, xrpl-validator) New amm_swap module: faithful port of rippled AMMHelpers under fixAMMv1_1/v1_2. A 16-significant-digit decimal Number core with per-op directed rounding (correctly rounded mul/div/add/sub/sqrt), swapAssetIn/swapAssetOut with the amendment's AMM-favoring rounding sequence, spot- quality gating vs the book (strictly-better + 1e-7 relative threshold), book-anchored sizing via changeSpotPriceQuality (citardauq …

Independence program

No independence record for UTC day 2026-07-19; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Selective-Disclosure Merkle Audit Trails** (audit) Add a new owner-reserved ledger object `ltAUDIT_COMMITMENT` (keylet::auditCommitment(owner, sequence)) holding an append-only Merkle Mountain Range ro
- **Time-Limited NFT Rental Escrow** (nft) Add a `tfRental` flag and `sfRentExpiration` (LedgerIndex) SField to `NFTokenEscrowCreate` (XLS-51). When set, `EscrowFinish` does not reassign the `N
- **Whitelisted On-Ledger Parameter Voting** (governance) Add a singleton-per-key, reserve-free ledger object `ltPARAMETER` (keylet::parameter(name), same footing as `Amendments` / `FeeSettings`) storing `sfPar

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-19 11:40	—	xrpl-validator	`150c432`	feat(xrpl-ledger): order-book quality keylet + OfferCreate placement — base camp
07-19 12:38	—	xrpl-validator	`7ee6dcd`	feat(xrpl-ledger): integer offer-crossing engine — the summit assault
07-19 13:34	—	xrpl-validator	`f5b87b2`	fix(xrpl-ledger): cancel-and-replace + funded-check starvation — OfferCreate 64%
07-19 14:43	—	xrpl-validator	`e6efe32`	feat(xrpl-ledger): IOU + cross-currency payment engine — second summit
07-19 15:58	—	xrpl-validator	`a4dc8d7`	feat(xrpl-ledger): CheckCash via shared transfer engine — IOU checks + tecPATH_PARTIAL semantics
07-19 16:29	—	xrpl-validator	`02508d9`	feat(xrpl-ledger): CheckCash via transfer engine + tec classification — corpus 90%
07-19 19:50	—	xrpl-validator	`28f0cb6`	fix(xrpl-ledger): path payments — quality gate only under tfLimitQuality, DeliverMin floor, book pre-state — corpus 90.3%
07-20 00:12	—	xrpl-validator	`18d4a00`	feat(xrpl-ledger): AMM swap engine — corpus 94.1%, the third summit

Times in UTC. A commit appears on the day it was committed to main or merged into it.